
MACHINE LEARNING

WORKSHOP

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2026

GROUP 630

Hand Exercises:

Consider the following six two-dimensional data points:

Point	Coordinate
A	(1, 1)
B	(2, 1)
C	(2, 3)
D	(6, 5)
E	(7, 6)
F	(8, 5)

Suppose we wish to cluster the data into two clusters. The initial cluster centers are $\mu_1 = (1, 2)$ and $\mu_2 = (7, 5)$

- Compute the distance from each data point to both cluster centers.
- Assign each point to the nearest cluster.
- Compute the updated cluster centers after the assignment step.
- Using the updated cluster centers, compute the new reconstruction error.

Task a We calculate the Euclidean distance, this can be done with the following formula:

$$d = \sqrt{(x_1 - \mu_x)^2 + (x_2 - \mu_y)^2}$$

Calculations was done in maple:

$$\begin{aligned}x_1 &:= 1 = 1 \\y_1 &:= 1 = 1 \\\mu_x &:= 1 = 1 \\\mu_y &:= 2 = 2 \\d_{(\mu_1)} &= \sqrt{(x_1 - \mu_x)^2 + (y_1 - \mu_y)^2} = d_{\mu_1} = 1\end{aligned}$$

Figure 1: Maple calculation

Point	Coordinate	Distance to μ_1 (D1)	Distance to μ_2 (D2)
A	(1, 1)	1	7.21
B	(2, 1)	1.41	6.40
C	(2, 3)	1.41	5.39
D	(6, 5)	5.83	1
E	(7, 6)	7.21	1
F	(8, 5)	7.62	1

Task b

- A: $D1 < D2$, point A belongs to Cluster 1;
- B: $D1 < D2$, point B belongs to Cluster 1;
- C: $D1 < D2$, point C belongs to Cluster 1;
- D: $D1 > D2$, point D belongs to Cluster 2;
- E: $D1 > D2$, point E belongs to Cluster 2;
- F: $D1 > D2$, point F belongs to Cluster 2;

So,

- Cluster 1: A, B, C;
- Cluster 2: D, E, F.

Task c To compute the updated cluster center, we take the mean of the x-coordinates and the mean of the y-coordinates for all points in each cluster

$$\mu_1 = \left(\frac{1+2+2}{3}, \frac{1+1+3}{3} \right) = (1.67, 1.67)$$

$$\mu_2 = \left(\frac{6+7+8}{3}, \frac{5+6+5}{3} \right) = (7, 5.33)$$

Task d

The reconstruction error is calculated as the sum of the squared distances between each point and its assigned cluster center.

First we use this formula, using the updated cluster centers. $(x_1 - \mu_x)^2 + (y_1 - \mu_y)^2$

For cluster 1:

A: $\|(1-5/3)^2 + (1-5/3)^2\| = 4/9$ B: $\|(2-5/3)^2 + (1-5/3)^2\| = 5/9$ C: $\|(2-5/3)^2 + (3-5/3)^2\| = 17/9$

$$E_1 = \frac{8}{9} + \frac{5}{9} + \frac{17}{9} = \frac{10}{3}$$

For cluster 2:

D, E, F

$$E_2 = \frac{10}{9} + \frac{4}{9} + \frac{10}{9} = \frac{8}{3}$$

Thus, the total reconstruction error is

$$E = E_1 + E_2 = \frac{10}{3} + \frac{8}{3} = 6.$$